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The impact of women's political representation on child health outcomes during 1990–2020: Evidence from a global dataset

Niharika Rustagi *, Sonia Akter

Lee Kuan Yew School of Public Policy, National University of Singapore, 469c Bukit Timah Road, 259772, Singapore

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ABSTRACT

This study investigates the impact of women's political representation on children's health outcomes from 1990 to 2020 using a global dataset comprising 162 countries. The child health outcomes studied are infant and neonatal mortality rates and vaccination coverage for measles and diphtheria, pertussis, and tetanus (DPT). We apply an event study and panel data instrumental-variable fixed-effects regression analysis for data analysis. The timing of the introduction of a gender quota in a country is used as an instrument to address endogeneity bias commonly associated with women's political participation. We find evidence of a positive impact of increasing women's representation in national legislatures on all child health outcomes studied. In particular, our study reports a significant negative effect of women's political participation on neonatal mortality which was not reported by any previous research. We also find evidence to support hypotheses related to several potential pathways that connect women's political representation with child health outcomes, including health expenditure, female labor force participation, skilled birth attendance, and adolescent fertility rate. Also novel in our findings is a heterogeneity analysis which reveals that reserved seat quota boosted child health outcomes more than candidate quota, with strong geographical variation in the relationship. Sub-Saharan Africa and South Asia benefited the most in terms of child health outcomes due to an increase in women's political representation among all regions. Our findings reinforce the significance of women's greater political empowerment for child health—a crucial development outcome—particularly in regions characterized by the poor quality of early childhood health conditions and survival rates.

1. Introduction

Gender equality is a fundamental development goal in its own right, but it also has positive externalities. For example, studies have shown the positive effects of gender equality on a wide range of child health, nutrition, food security, and education outcomes (Ewerling et al., 2020). A growing body of literature also focuses on understanding the impact of women's greater political representation on policy choices and outcomes for women and children (Hessami and da Fonseca, 2020).

Although theory predicts that politicians' personal characteristics do not affect policy choices (Downs, 1957), empirical evidence suggests that policy choices vary across political representatives' gender. Theoretically, political leaders cater to the median voter due to concerns about re-election, making the gender of the policymaker irrelevant. However, empirical evidence shows that politicians have preferences over policies, which vary across the gender of the policy maker. For example, female politicians are more likely than male politicians to support public health improvements directed toward female constituents as they disproportionately bear the burden of weak reproductive and child health services (Cutler et al., 2006). In addition, women primarily bear the burden of care for the sick, witness the death of their children, undergo childbirth, and face the risk of mortality during childbirth (Bhalotra and Van Soest, 2008). In India, female political leaders were more likely to support a higher provisioning of public goods (Beaman, 2007) and policies that reflect women's needs such as drinking water and maternity pension than their male counterparts (Chattopadhyay and Duflo, 2004). Furthermore, studies show that in villages with female leaders, girls spent less time on household chores, reducing the gender gap in school attendance and educational attainment, than in villages with male leaders (Beaman, 2007; Beaman et al., 2012).

One of the explanations for gender differences in development

Abbreviations: DPT, diphtheria, pertussis, and tetanus; IMR, Infant Mortality Rate; NMR, Neonatal Mortality Rate.

* Corresponding author.
E-mail addresses: niharika.rustagi@u.nus.edu (N. Rustagi), sonia.akter@nus.edu.sg (S. Akter).

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outcomes is gender disparities in policy preferences (Chattopadhyay and Duflo, 2004; Eckel and Grossman, 1998). Another explanation lies in a gender difference in rent-seeking behavior. In developed and developing countries, women's greater political representation is associated with lower corruption (Dollar et al., 2001; Swamy et al., 2001). In Brazil, Brollo and Troiano (2016) found that female politicians are less likely to accept bribes and condone corruption than male politicians. Using data on mayoral elections, the authors also found a negative correlation between women's representation and corruption related to random audits of local administrations. Their findings suggest that female mayors receive less patronage and are less susceptible to private financial interests than male mayors.

This study investigates the causal relationship between women's political representation and child health outcomes in 162 countries from 1990 to 2020. This is the first study of its kind undertaken on a global scale using 30 years of longitudinal data. Previous studies examining this relationship used cross-sectional data for 51 low- and middle-income countries and longitudinal data for 20 countries from 2003 to 2011 (Quamruzzaman and Lange, 2016). The studies reported that female political representation was negatively correlated with infant mortality and measles vaccination status. In another cross-country study, Swiss et al. (2012) used a random-effects panel regression model to study the influence of women's representation on child health across 102 developing countries from 1980 to 2005. They found women's representation to be positively correlated with an improvement in immunization and child survival rate.

Quamruzzaman and Lange (2016) and Swiss et al. (2012) found strong positive correlations between women's political representation and child health outcomes. However, the endogeneity bias associated with women's political participation needs to be addressed explicitly to prove causality. This bias could arise from two sources: (1) reverse causality, i.e., healthy children reduce women's unpaid care work, thus increasing women's opportunity to participate in politics; (2) omitted variable bias, i.e., in countries with increasing women's political participation, children's health outcomes simultaneously improve due to unobserved factors that positively affect both political participation and children's health outcomes.

We address the endogeneity bias using an event study and a quasi-experimental technique, i.e., instrumental variable analysis. Additionally, we investigate several indicators that operationalize child health: infant mortality rate, neonatal mortality rate, and measles and DPT vaccination coverage for the 12–23 month age group. We also explore infant mortality rates by sex and variation in the link between women's political representation and child health outcomes by gender quota type (i.e., reserved seat versus candidate quota) and geographical region.

For the instrumental variable analysis, we exploit random variation in the timing of the introduction of a gender quota in a country as an instrument. Gender quotas refer to mandates to improve women's political representation. Despite not being very common before the 1990s, more than 130 countries now have provisions in their constitution, electoral laws, or voluntary party quotas requiring the inclusion of women as legislators or candidates (Hughes et al., 2019). This reform was driven by an international movement to improve women's political representation, achieving major success in 1995 at the United Nations Fourth World Congress on Women in Beijing (Fallon et al., 2012). Given the rapid pace of quota adoption worldwide following the Beijing World Congress, studies on the causes and consequences of quotas have become some of the fastest-growing works of literature in gender and politics research (Hughes et al., 2017b). Two broad types of gender quotas are adopted by countries: (1) the reservation of seats in legislative bodies, usually stipulated by national law requiring a certain percentage of seats to be occupied by women; (2) the candidate quota, requiring a certain percentage of women to occupy places on party lists, mostly in proportional representation electoral systems (Clayton, 2021). Reserved seats are the most direct, effective way to boost women's political representation because the candidate quota does not guarantee women's political representation in the legislature.

Our contribution to the literature on the political economy of health is manifold. In addition to using a global sample of 162 countries and investigating a comprehensive range of variables measuring child health outcomes, we report several mechanisms mediating the effect of women's political participation. More specifically, we use four variables—female labor force participation rate, health expenditure as a proportion of gross domestic product (GDP), skilled birth attendance and adolescent fertility rate—to shed light on possible mechanisms mediating the relationship between the treatment and outcome variables. With the exception of health expenditure, these mechanisms have not been discussed in literature as possible pathways mediating the relationship between women's representation and child health outcomes. We also address the endogeneity associated with women's political representation that potentially biased the findings of the existing studies. Our study is the first to report a significant negative effect of women's political participation on neonatal mortality. Another important contribution of our study is heterogeneity analysis across quota type and regions. Our results show a more pronounced effect of women's representation on child health outcomes under reserved seat quotas than candidate quotas along with a considerable difference in the effect of women's political participation on child health outcomes in Sub-Saharan Africa and South Asia relative to the rest of the world. Our main results are robust to several econometric specifications and all data analysis methods.

2. Literature

The political economy of health, especially focusing on the gender of politicians, has gained much attention from researchers in recent decades. Bhalotra and Clots-Figueras (2014) found that constituencies electing a higher proportion of women in India saw higher investments in public health infrastructure (provisioning of antenatal and childhood services), significantly reducing neonatal mortality. Similarly, Macmillan et al. (2018) found that female representation negatively correlated with child mortality in weak national democratic contexts, e.g., Haiti, Libya, and Somalia.

Research shows that female legislators are more likely to make budgetary allocations and enact policies that promote healthcare and address issues related to women's and children's welfare, education, social services, and social protection spending than their male counterparts (Funk and Philips, 2019). Research in the USA shows that policies supporting Medicaid, infant and maternal care programs, access to abortion, and subsidized family planning are associated with reduced infant mortality and improved child and maternal well-being (Bailey et al., 2013). However, the literature tends to focus on analyzing the context of certain countries or sub-national levels, while the analysis of broader country-level patterns across regions is scarce (Cunial, 2021). A global approach contrasts with case analysis, allowing researchers to test the insights from case study research and explore regional and global effects.

The literature around gender quotas is burgeoning, but evidence of their effects remains mixed. Studies have reported that quota adoption either increases the substantive representation of women in legislature (Clayton et al., 2016) or gender inegalitarian outcomes (Brulé, 2020). Quotas are expected to influence policy in two ways (Clayton, 2021): (1) sending cues to legislation makers, affecting overall change in legislative behavior among men and women; (2) changing aggregate policy preferences and priorities, increasing women's ability to influence legislative decisions collectively.

All 189 participating states signed the Beijing Declaration and Platform for Action in 1995, asking governments to take proactive measures to ensure women's equal participation in power and decision-making structures. As a result, local and international pressure increased to adopt gender quotas since 1995. Most international development agencies view quotas as crucial for improving women's presence and

power within national political institutions. However, the spread of quotas has not been even across all regions worldwide (see Fig. 1), with the highest adoption in countries across Sub-Saharan Africa, followed by Europe, Central Asia, and South America. These differences could be related to the support of quotas by political elites, rooted in the strategic advantages offered by quotas (Bush and Zetterberg, 2020). Research on European democracies has shown that elites favor quotas due to inter-party and intraparty competition: parties might attempt to attract new groups of voters by signaling that they are progressive (Murray et al., 2012; Weeks, 2018). Men incumbents may also try to increase their chances of re-election by facing inexperienced women candidates with little voter support (Fréchette et al., 2008), and male party leaders may believe that quotas give them more control over candidate selection (Weeks, 2018). Strategic reasons may also explain the uptake of quotas in non-democratic countries. These institutions help enhance international reputations for democracy, deflecting external pressure to democratize in the post-Cold War environment. For example, in a cross-national study of quota adoption in the developing world, Bush (2011) posits that countries signal their commitment to democracy by adopting quotas.

The role of the international community was central in the adoption of gender quotas (Norris and Dahlerup, 2015). International organizations created a conducive envrionemnt for quota adoption by establishing conventions, treaties and agreements which shaped global norms, including strengthening the principles of gender equality and women's rights (Bush, 2011). After the United Nations Fourth World Congress on Women in Beijing in 1995, the proportion of women in national parliaments increased from 10% in 1995 to around 22% by 2015. The current geographic spread of quotas can be seen in Fig. 2. This was majorly driven by the spread of gender quota policies across many regions of the world. They were a popular institutional reform designed to fast-track greater inclusivity and diversity in parliaments (Krook, 2010). This thrust towards gender equality in elected office was driven by the 2000–2015 MDGs. Adoption of quotas was brought into the limelight again by shifting focus from supply-side factors like women's apparent lack of political interest to pointing towards political institutions on the demand side along with political parties for their lack of inclusiveness. Different kinds of quotas have been adopted by both democratic as well as hybrid regimes and electoral autocracies (Fallon et al., 2012).

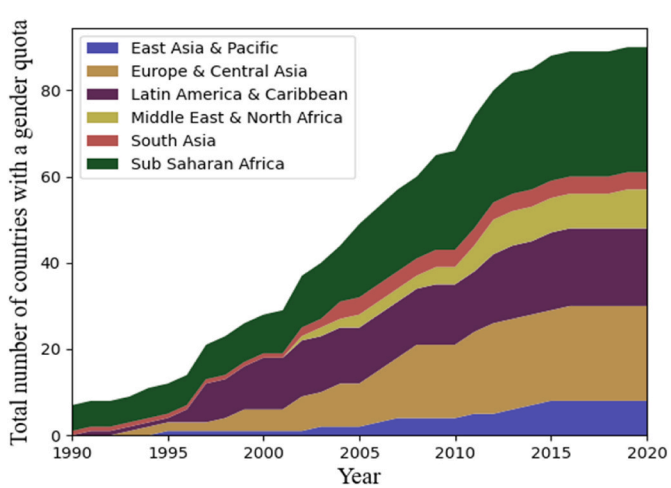

Fig. 1. Adoption of electoral gender quotas by region.
Data sources: Dahlerup (2005), Global Database of Quotas for Women (Inter-Parliamentary Union, 2015), and Hughes et al. (2017a).

3. Methods

3.1. Data sources and measures

3.1.1. Outcome variables

We operationalize child health using four outcome variables: (1) *Infant Mortality Rate* (IMR), the number of deaths occurring among infants under one year of age per 1,000 live births (included in the models as rate per 1,000); (2) *Neonatal Mortality Rate* (NMR), the number of infant deaths in the first 28 days of life, expressed per 1,000 live births; (3) *Measles Vaccination Coverage*, the percentage of children aged 12–23 months who received measles vaccination; (4) *DPT Vaccination Coverage*, the percentage of children aged 12–23 months who received the diphtheria, pertussis, and tetanus (DPT) vaccination. All definitions and data for the outcome variables were collected from the World Development Indicator (World Bank, 2020) database.

3.1.2. Treatment variable

The data on women's political representation is defined as the percentage of parliamentary seats women hold in a single or lower chamber of the national parliament (Inter-Parliamentary Union, 2015). We use two sources for compiling annual measures of the proportion of seats occupied by women in parliament: (1) WDI Indicator (World Bank, 2020), providing data on the percentage of seats held by women in the national parliament from 1997 to 2020; (2) Interuniversity Consortium for Political and Social Research dataset (Paxton et al., 2008), providing data on parliamentary seats held by women from 1990 to 1996.

3.1.3. Gender quota

The data for country-specific adoption of gender quotas was collated from Dahlerup (2005) and updated using the Global Database of Quotas for Women (Inter-Parliamentary Union database), along with the Quota Adoption and Reform Over Time dataset compiled by Hughes et al. (2017a). The first dataset covers data on gender quota adoption until 2005, while the other two cover different countries post-2005. The presence or absence of a gender quota is used as an instrument. It takes a value of 1 for all years post-adoption of the quota for adopting countries and 0 otherwise.

3.1.4. Control variables

We include two control variables for analysis: (1) natural logarithm of GDP per capita in 2010 measured in constant international dollars (PPP adjusted), as income correlates with health outcomes and women's political representation; (2) quality of democracy (Bhalotra et al., 2021; Quamruzzaman and Lange, 2016). Quotas were adopted by democratic, authoritarian, and hybrid regimes (Hughes et al., 2015), but the expected direction of the relationship between quotas and policy outcomes varies across cases (Baldez, 2006). This data was taken from V-DEM (Varieties of Democracy) (Coppedge et al., 2015) and POLITY databases. The extent of democracy is measured on a scale of 0–1 for the V-DEM database and 1 to 10 for the POLITY score. In both cases, 0 and 1 refer to the least democratic regime or the lowest quality of democracy, and 1 and 10 are the highest, respectively.

3.2. Empirical strategy

Combining the variables discussed in the previous section, we constructed a longitudinal dataset spanning 162 countries over 30 years, i.e., from 1990 to 2020. For investigating the impacts of the passage of gender quotas without restricting the timing of effects, the following event study[1] was carried out:

[1] The event study regression specification was developed following Bhalotra et al. (2021).

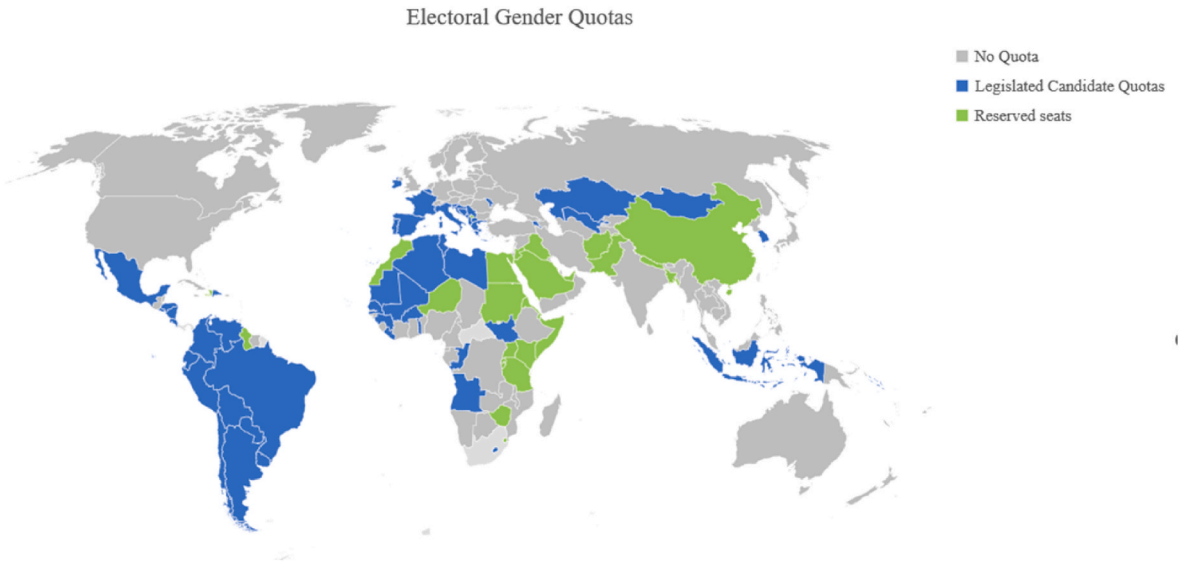

Fig. 2. Map of countries that have adopted electoral gender quotas.
Data sources: Dahlerup (2005), Global Database of Quotas for Women (Inter-Parliamentary Union, 2015), and Hughes et al. (2017a).

$$Y_{ct} = \alpha + \sum_{l=2}^{10+} \beta_l^{lead} Quota_c \times 1\{lead_t = 1\} + \sum_{k=0}^{10+} \beta_k^{lag} Quota_c \times 1\{lag_t = k\} + X_{ct}\gamma + \mu_t + \varphi_c + \varepsilon_{ct} \quad (1)$$

In eq. (1) *Yct* denotes a child health outcome. The variation in health outcome is across country *c* and year *t*. $Quota_c$ is 1 if a country ever adopted a gender quota, it interacted with a full set of leads and lags with respect to the year the quota was adopted. The first lead is omitted as the base category. The β^{lag} coefficients capture the impacts of interest

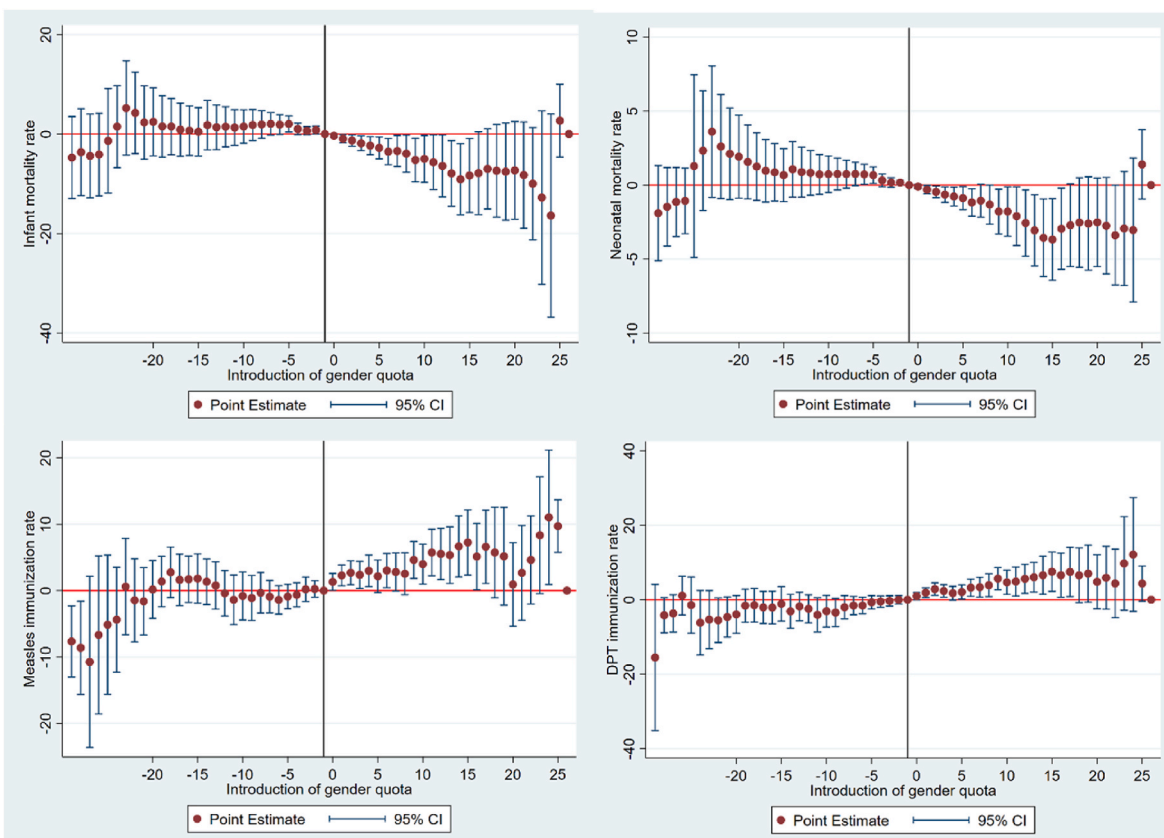

Fig. 3. Event studies for child health outcomes.
Notes: Point estimates of the lag and lead terms in the event study specification are presented, along with their 95% confidence intervals. Estimates are conditional on country and year fixed effects, the natural logarithm of per capita GDP, and democracy index. Robust standard errors are clustered at the country level. The omitted base category is taken as 1 year prior to the reform, indicated by the solid vertical line.
DPT, diphtheria, pertussis, and tetanus

and the β^{lead} coefficients test the identifying assumption of no differential pre-trends. We add country and year fixed effects (φ_c and μ_t respectively), and cluster standard errors at the country level. When applying the event study methodology to our study, we found no evidence of differential pre-trends before quota adoption in any of the outcome variables as seen from Fig. 3.

Next, we conducted two-stage least squares regression for addressing potential endogeneity concerns. The longitudinal dataset allows us to estimate the effect of women's political representation over time, and the gender quota instrument helps combat concerns of reverse causality and omitted variable bias.

The following equations outline the empirical specifications used for the instrumental variable fixed effects (IVFE) model:

$$M_{ct} = \alpha 0 + \alpha 1 Zct + \mu_t + \varnothing_c + \eta ct \quad (2)$$

$$Y_{ct} = \beta_0 + \beta_1 \widehat{M_{ct}} + \mu_t + \varnothing_c + \varepsilon_{ct} \quad (3)$$

Equation (2) denotes the first stage fixed-effects regression model of the proportion of women explained by the 'excluded instrument' of gender quotas (Z_{ct}). It equals 1 for all years after quota adoption for adopting countries and 0 for pre-adoption years and non-adopting countries. M_{ct} is the proportion of women in parliament for country 'c' in year 't'. The year and country fixed-effects are denoted by μ_t, $\varnothing_c$. Equation (3) is the second stage fixed-effects regression of child health outcomes (Y_{ct}) explained by fitted values (M_{ct}) of the proportion of women in parliament from the first stage. Z_{ct} is excluded from the second stage and is thus referred to as the excluded instrument.

To check the validity of the exchangeability assumption of the instrument, we present the distribution of key socio-economic variables (i.e. quality of democracy, per capita income and overall labor force participation rate) of all countries in our sample. The results are presented in Appendix 2. No significant difference in pre-treatment levels between adopters and non-adopters of the gender quota policy was observed. These findings suggest that the decision to implement the gender quota introduction timing was quasi-random.

The possibility of the violation of exclusion restriction poses a threat to instrument validity. The exclusion restriction would be violated if the quota legislation influenced child health outcomes directly, rather than or in addition to influencing the outcomes of interest through women's political participation. For example, the introduction of gender quotas could lead to 'signal' or 'interpretive' effects (Zetterberg, 2009; Pierson, 1993) by affecting the attitudes and involvement of the citizens of under-represented groups. However, empirical evidence does not support the theory. In a randomized policy experiment, Clayton (2015) found no effect of gender quotas on redefining the individual-level attitudes and behavior. As mentioned before, quotas are expected to influence policy in two ways (Clayton, 2021): (1) sending cues to legislation makers, affecting overall change in legislative behavior among men and women; (2) changing aggregate policy preferences and priorities, increasing women's ability to influence legislative decisions collectively. The treatment variable of our study, i.e., the proportion of women in parliament captures both of these pathways. Therefore, the presence of quota is only likely to affect child health outcomes indirectly through women's political participation.

4. Results

4.1. Descriptive statistics

Table 1 provides the descriptive statistics for all the variables used in the analysis. Notably, males had a higher average IMR than females. The mean percentage of women in parliament is 15.6%, ranging from 0 to 63.75%. The mean percentages for measles and DPT vaccine coverage are similar at 83.6% and 84.5%, respectively, with less variation than IMR. For the mechanism variables, the adolescent fertility rate has the highest variation.

4.2. Regression results

The regression estimates for Equation (1) can be seen in Fig. 3 for all four outcome variables. In all four figures, a discrete jump can be seen in the health outcomes in one to two years after quota implementation. The figures show a clear break in coefficients, with mortality rates falling and immunization rates rising faster in countries with quota implementation. The lead coefficients subdue the endogeneity concern of adopting gender quota policy.

Estimates for Equations (2) and (3) are in Table 2. The first-stage regression coefficient reveals a positive correlation between gender quotas and women's representation in parliament, with the presence of gender quotas increasing the likelihood of women being in parliament by 4.8%. The F-statistic is 15.66 and statistically significant at the 1% level. The results are not sensitive to including population weights as an additional control variable. The Hausman test shows whether the panel FE estimates statistically differ from the IVFE estimate at $p < 0.05$. In cases of statistical difference, the treatment variable (proportion of women in parliament) is endogenous for all outcome variables. Thus, the IVFE estimate is better for estimating the true effect of women's political representation than the panel FE model. The panel FE regression results are in Appendix 1.

The regression coefficient (Table 2, Column 1) suggests that a one percentage point increase in women's political representation reduces infant mortality by approximately 0.9 per 1,000 live births ($p < 0.05$). This 0.7% reduction compared to the baseline is much greater than the 0.1% reported by Quamruzzaman and Lange (2016). This effect is substantial given our sample's average infant mortality rate (3.36%). For disaggregated effects, a similar reduction of 0.7% was observed for both male and female IMR (Table 2, Columns 2 and 3). Like infant mortality, the effect of women's political representation in parliament negatively affects neonatal mortality rates. A one percentage point

Table 1
Descriptive statistics.

Variable	Observations	Mean	Std. Dev.	Min	Max
% of women in parliament	5,319	15.67	11.27	0.00	63.75
Infant mortality rate (IMR) (mortality rate, infant (per 1,000 live births)	5,647	33.60	31.25	1.50	176.50
Female IMR (mortality rate, infant, female (per 1,000 live births))	5,647	30.53	28.81	1.40	163.20
Male IMR (mortality rate, infant, male (per 1,000 live births))	5,647	36.52	33.60	1.60	189.20
Neonatal mortality rate (mortality rate, newborns in first 28 days (per 1,000 live births))	5,647	18.08	14.37	0.80	74.30
Immunization—Measles (% of children aged 12–23 months)	5,512	83.60	16.70	8.00	99.00
Immunization—DPT (% of children aged 12–23 months)	5,520	84.50	17.07	10.00	99.00
Labor force participation rate, female (% of female population age 15–64)	5,284	40.27	9.63	7.80	56.00
Health expenditure (% of GDP)	4,490	5.65	2.88	0.55	28.05
Births attended by skilled health staff (% of total)	2,554	88.95	20.35	5.00	100.00
Adolescent fertility rate (births per 1,000 women aged 15–19 years)	5,377	60.63	47.88	0.28	223.69
Gross Domestic Product per capita in 2010 US$	5,517	13,183.87	21,224.74	164.33	209,224.50
Democracy index (0–1)	4,996	0.41	0.27	0.01	0.89

Table 2
Impact of women's political representation on children's health outcomes (IVFE regression results).

	(1)	(2)	(3)	(4)	(5)	(6)
	IMR	Female IMR	Male IMR	Neonatal mortality rate	Measles immunization rate	DPT immunization rate
% women in parliament	−0.877**	−0.797**	−0.953**	−0.318**	0.610**	0.840**
	(0.367)	(0.347)	(0.387)	(0.145)	(0.260)	(0.290)
Ln of per capita GDP (in 2010 US$)	−7.451***	−6.758***	−8.131***	−4.152***	−2.500	−2.570
	(2.671)	(2.532)	(2.813)	(1.195)	(3.720)	(3.140)
Democracy index	−10.600	−10.040	−11.150	−4.022	3.480	3.910
	(7.015)	(6.602)	(7.419)	(2.611)	(6.010)	(6.350)
Constant	119.200***	108.600***	129.500***	61.530***	89.960***	90.820***
	(21.150)	(20.010)	(22.320)	(9.733)	(29.890)	(25.250)
First stage F-statistics	15.660				16.010	
First stage regression coefficient	4.790***				4.830*** (0.930)	
	(0.940)					
Country FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
R-squared	0.460	0.450	0.460	0.530	0.140	0.040
Observations	4,279	4,279	4,279	4,279	4,265	4,265
Number of countries	162	162	162	162	162	162
[a]Hausman test at $p < 0.05$ $\gamma_{FE} = \beta 1_{IV}$	Reject	Reject	Reject	Reject	Reject	Reject

Notes: IVFE stands for instrumental variable fixed effects. Robust standard errors clustered at the country level in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.
IMR, Infant mortality rate.
DPT, diphtheria, pertussis, and tetanus.
[a] Hausman test determines whether the treatment variable is endogenous.
Thus, rejecting the null hypothesis indicates that this treatment variable is likely to be endogenous.

increase in women's political representation reduces the average neonatal mortality rate by 0.3 points per 1,000 children, in contrast with Macmillan et al. (2018), who reported no significant effect of women's political representation on neonatal deaths. We also found a significant positive effect of women's political representation on measles and DPT vaccine coverage (Table 2, Columns 5 and 6), consistent in magnitude with Quamruzzaman and Lange (2016) for measles vaccination.

Panel FE model estimates (Appendix 1) was not significantly associated with women's representation in parliament with any child health outcome variable. In line with expectation, GDP per capita is negatively associated with infant and neonatal mortality rates. Similarly, improvement in the institutional variable of democracy was negatively correlated with mortality rates in the FE model. The results were not sensitive to the inclusion of country and year FE.

4.3. Heterogeneity analysis

We carried out several heterogeneity analyses as part of the data analysis. Firstly, we tested whether geographical variation exists in the effect of women's political representation on child health outcomes (Table 3). According to the World Bank (2017) classification, there are seven geographical regions in the world: East Asia and Pacific, Europe and Central Asia, Latin America and the Caribbean, the Middle East and North Africa, North America, South Asia, and Sub-Saharan Africa. The regression coefficients presented in Fig. 4 indicates that increasing the proportion of women in the parliament did not significantly impact infant and neonatal mortality rates and child immunization rates for North America and European and Central Asian countries. All other regions had statistically significant regression coefficients with similar effect sizes, except for the neonatal mortality rate (a weak negative correlation). We also tested whether variation existed for quota types. Fig. 5 shows the regression coefficients for countries with candidate quotas and reserved seats. For all four mortality rates, statistically significant associations occurred for reserved seat quotas, with effect sizes consistent with the main results (Table 4). A more strongly significant effect of women's representation on female IMR was observed with no change in effect size, compared to the overall sample. In contrast, no impact of candidate quotas was observed for any mortality outcome, which could stem from the smaller impact of candidate quotas on women's share in parliament.

4.4. Mechanism checks

We explored the underlying mechanisms mediating the impact of female political representation on child health outcomes. Table 5 reports the estimates for four hypothesized mediating variables. Columns 1 and 2 show that women's representation in the parliament was positively associated with the female labor force participation rate and health

Table 3
Heterogeneity analysis by region (IVFE regression results).

	IMR		Neonatal mortality rate		Measles immunization rate		DPT immunization rate	
	All other regions	North America & Europe	All other regions	North America & Europe	All other regions	North America & Europe	All other regions	North America & Europe
% women in parliament	−0.830**	0.070	−0.310*	0.040	0.790**	−0.360	0.870**	0.110
	(0.420)	(0.400)	(0.160)	(0.170)	(0.310)	(0.430)	(0.340)	(0.310)
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y	Y	Y
R-squared	0.55	0.56	0.58	0.63	0.09	0.19	0.11	0.14
Observations	2945	1334	2945	1334	2943	1322	2943	1322

Notes: IVFE stands for instrumental variable fixed effects. Robust standard errors clustered at the country level in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.
IMR, Infant mortality rate.
DPT, diphtheria, pertussis, and tetanus.

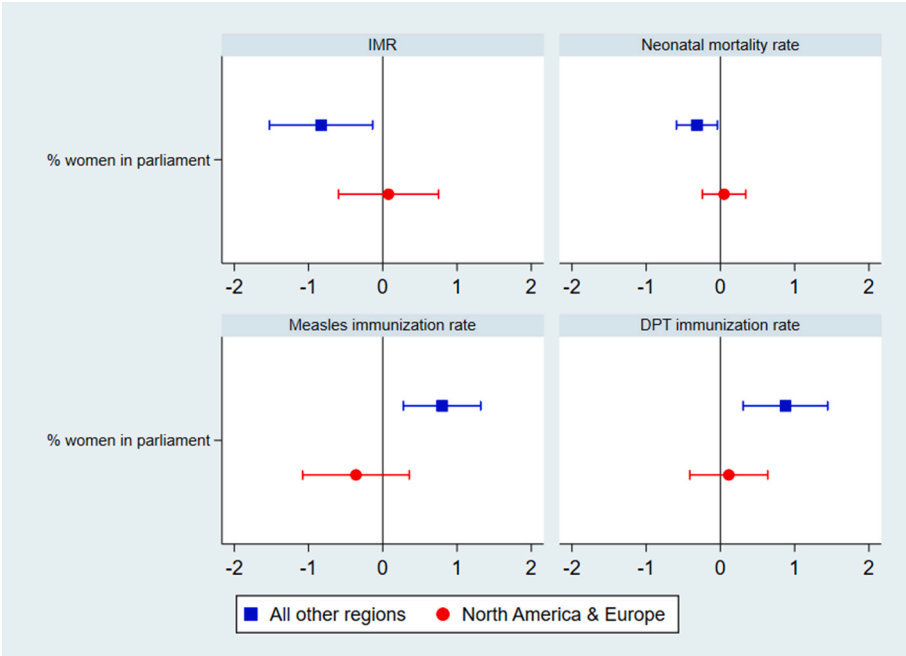

Fig. 4. Heterogeneity analysis by region (IVFE regression coefficients and 90% confidence interval). Notes: Regression coefficients are shown for the heterogeneity analysis for different regions. All regression models include control variables (i.e., democracy index and GDP) and country and time FE. IVFE stands for instrument variable fixed effects. Robust standard errors are clustered at the country level.
IMR, Infant mortality rate.
DPT, diphtheria, pertussis, and tetanus.

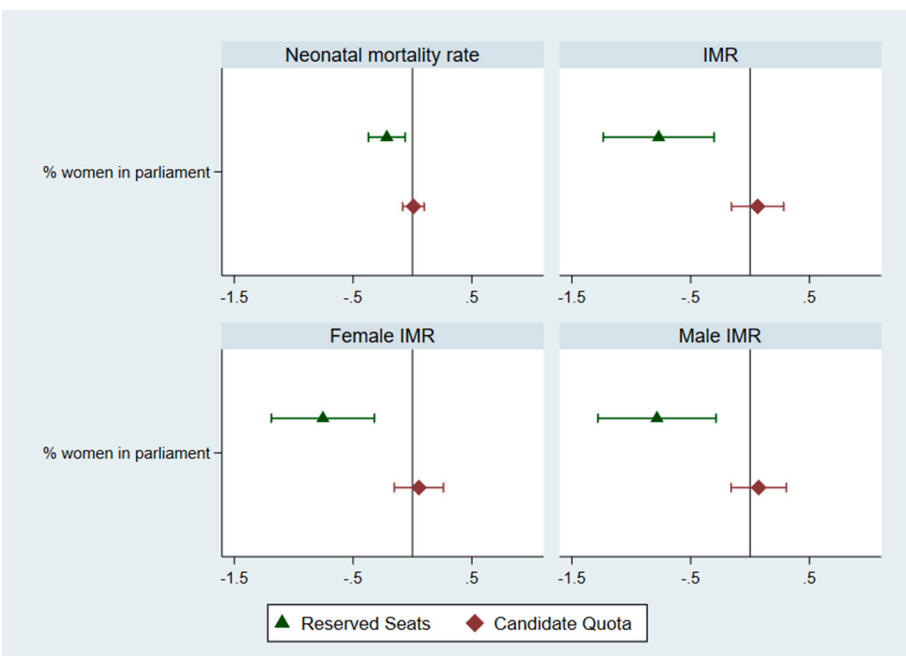

Fig. 5. Heterogeneity analysis for the impact of gender quota types on child health outcomes (FE regression coefficients and 90% confidence intervals). Notes: Regression coefficients are shown for the heterogeneity analysis for different quota types. Each regression includes time FE. Robust standard errors are clustered at the country level.
IMR, Infant mortality rate.

Table 4
Impact of gender quota types on child health outcomes (OLS regression results).

	IMR		Female IMR		Male IMR		Neonatal mortality rate	
	Reserved seats	Candidate quota	Reserved seats	Candidate quota	Reserved seats	Candidate quota	Reserved seats	Candidate quota
% women in parliament	−0.760**	0.060	−0.754***	0.050	−0.783**	0.070	−0.210**	0.008
	(0.260)	(0.130)	(0.251)	(0.120)	(0.287)	(0.130)	(0.080)	(0.050)
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y	Y	Y
R-squared	0.77	0.57	0.77	0.56	0.77	0.58	0.69	0.63
Observations	496	1589	496	1589	496	1589	496	1589

Notes: Robust standard errors clustered at the country level in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.
IMR, Infant mortality rate.

Table 5
Mechanism check (OLS regression results).

	(1)	(2)	(3)	(4)
	Female labour force participation rate	Health expenditure	Adolescent fertility rate	Attended births
% women in parliament	0.100***	0.090***	−0.530***	0.180***
	(0.140)	(0.010)	(0.070)	(0.060)
Ln of per capita GDP (in 2010 US$)	0.360	1.580***	−9.900***	9.320***
	(0.520)	(0.400)	(2.380)	(1.900)
Democracy index	0.250	2.120**	−13.200*	17.040**
	(1.060)	(0.980)	(7.180)	(8.640)
Country FE	Y	Y	Y	Y
Time FE	Y	Y	Y	Y
R-squared	0.210	0.310	0.380	0.230
Observations	4,251	3,602	4,279	2,958
Number of countries	161	161	162	157

Notes: Robust standard errors clustered at the country level in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.

expenditure as a proportion of GDP. As skilled birth attendance rate was available only for each country in a subsample of years, we analyzed this hypothesized relationship using the subsample. Column 3 shows increased women's political participation associated with a reduction in the adolescent fertility rate. Column 4 shows a positive association between women's political participation and births attended by a skilled staff. The coefficients of all mechanisms variables were statistically significant at the 1% level.

4.5. Robustness checks

For the first robustness check, we substituted the instrument (i.e., the year of gender quota introduction) with the year of a country's first national election post-adoption of the gender quota. Thus, Equation (2) was re-estimated with the new instrument. The findings are reported in Appendix 3. This robustness check suffers from low power due to limited observations as the information on the first election post-adoption of gender quota was available only for a part of the sample. The results of the re-estimated models are consistent with the results presented in Table 2, with a slight increase in the coefficient values and significance for all outcome variables.

The second robustness check is comprised of a different measurement for the quality of democracy, given its significant relationship with child health outcomes at the country level. In addition to the V-DEM database, we used data from the Freedom House Index for Quality of Democracy (see Appendix 4) For the alternate measurement of the quality of democracy, the results are similar to the main results reported in Table 2 and consistent across the subsample, with statistically significant negative effects for infant and neonatal mortality rates and positive effects for child immunization rates with an increase in women's political participation. The relationship between women's political participation and mortality outcomes remains negative but statistically weaker than the main results, while the positive effect for immunization outcomes remains statistically significant at the 5% level.

5. Discussion

In this study, we examined the causal effect of women's political representation on child health outcomes using an event study style regression along with an instrumental variable regression analysis that used the year of gender quota introduction in a country as an instrument. Additionally, we studied the sex-disaggregated effect for a subset of these outcomes and the heterogeneity of the effect by gender quota type and regions. Our findings reveal a positive effect of women's political representation on four child health outcomes, i.e., infant and neonatal mortality rates and measles and DPT vaccine coverage. Compared to the results of Quamruzzaman and Lange (2016), we found a higher effect size of women's share in the parliament on the infant mortality rate. The difference of 0.6 percentage points is substantial given the average infant mortality rate in our sample (3.36%). This difference in magnitude could be due to the instrumental variable since our analysis estimates the Local Average Treatment Effect (LATE) rather than the Average Treatment Effect (ATE) estimated by other correlational studies, which could also explain the significant effect of women's political participation on neonatal mortality rate in our analysis. Macmillan et al. (2018) did not find a significant association between female political representation and neonatal mortality rate. These differences signify the importance of addressing endogeneity bias; if left unaddressed, the role of women's political participation in improving health outcomes for children could be underestimated.

The mechanism analysis reported in our study reveals new pathways through which enhanced political participation of women influences child health outcomes, including a reduction in adolescent fertility rate. The risk of higher neonatal mortality rates is correlated with adolescent motherhood is an important mechanism (Neal et al., 2018). In addition, women's increased political participation in India was associated with a reduction in child marriage (Castilla, 2018), a consequence of the increased aspirations of adolescent girls post-exposure to women political leaders (Beaman et al., 2012). Villages with female leaders also had girls more likely to be a part of the workforce and less likely to want to marry before 18 years of age (Beaman et al., 2012).

Another crucial mechanism reported as part of our analysis is women's labor force participation rate. Sadia (2020) reported that women's presence in the state legislature in India increased the probability of women's participation in wage employment through a dual policy and role model effect relative to their male counterparts. Subsequently, women with a greater economic empowerment were more likely to have fewer children and more likely to access healthcare services (Abreha and Zereyesus, 2021) than less empowered women, improving children's nutritional and health outcomes (Shroff et al., 2011).

Consistent with Quamruzzaman and Lange (2016), we found that health expenditure significantly mediated the relationship between women's political representation and infant and neonatal mortality rates and vaccination coverage, indicating the allocation of increased resources to health services focusing on children's welfare. This supports the channel of women legislators making budgetary allocations and enacting policies that promote healthcare and children's issues (Funk and Philips, 2019), resulting in improved child health outcomes. The importance of this channel is supported by the significant association between women's economic empowerment and the presence of skilled attendants during childbirth (Khatiwada et al., 2020). Reductions in neonatal mortality also require access to skilled attendance during labor and delivery (Edward et al., 2020; Fotso et al., 2020).

All four mechanisms point to improved economic and health indicators relating to women's empowerment, thus representing women's interests and supporting the argument of women legislators focusing on and addressing issues that traditionally or by nature primarily benefit women (Funk and Philips, 2019). However, the results of the mechanism analysis must be treated cautiously as they report correlations not causation.

We observed variation in the effect of women's representation across different regions, with significant effects on all child health outcomes for the regions of East Asia, Latin America, North Africa, South Asia, and Sub-Saharan Africa, consistent with the disparity in gender equality across different regions (Carmichael et al., 2014). According to the Human Development Report (UNDP, 2015), Sub-Saharan Africa, South Asia, and Latin America have greater gender inequality in reproductive health, political empowerment, and economic status than North America and European countries. The mechanism analysis, discussed above,

lends credence to the spatial distribution of health services as an important determinant of child survival (Simmons et al., 2021). Historically, gender inequalities in political rights, health status, and socioeconomic standing have been high across geographical regions (Carmichael et al., 2014) and remain significant. Even though Europe (including Eastern Europe) has performed best, it has not achieved complete gender equality. Women's weak political rights and status have kept the Middle East and Sub-Saharan Africa lagging in gender equality. South and Southeast Asia have not performed well due to a skewed sex ratio at birth.

In addition to variation across regions, the role of different gender quota types on child health outcomes is an important finding and contribution of our study. We found a significant effect of women's representation in parliament for all child health outcomes under reserved seat quotas, but no significant effect under candidate quotas. This is possibly because candidate quotas do not necessarily increase women's participation in parliament (Bhalotra et al., 2021). In addition, women elected through candidate quotas may have less influence than women in reserved seats. Proportional systems give party gatekeepers significant control (often absolute) over candidate selection, which may cause party leaders to choose women they think will toe the party line on policy issues (Clayton, 2021). The literature offers explanations for the differential effect of quotas yet Figs. 1 and 2 show that different geographical regions have adopted different types of quotas, making it difficult to isolate the effect of region from quota type, warranting further research.

One limitation of this study is that the analysis was conducted on an unbalanced panel. Limiting the analysis to countries with no missing data would have limited the sample size for multivariate analysis. Specifically, delineating the mechanisms mediating the effect of gender quotas on child health outcomes requires a strongly balanced panel with information on all covariates present for all countries. Future studies should examine the different mechanisms operating under different kinds of quotas and investigate the lags in improving health outcomes post-adoption of gender quotas. It would also be interesting to study whether a higher number of women in parliament affects the attitudes of all members or only women representatives on issues related to children, specifically educational outcomes.

6. Conclusion

Our findings shows that the increased presence of women in legislative bodies improves children's health outcomes. These results consider the endogeneity involved in studying the relationship between women's political representation with child health outcomes highlighting the role of gender quotas. Over time, the number of countries adopting this policy instrument to improve women's role in policy-making has increased. This study shows the merit of adopting this policy, specifically for reserved seat quotas. Along with exploring the geographical heterogeneity and variation in types of gender quota, we also highlight how increased women's representation in parliament translates into better health outcomes. This research warrants further investigation into the mechanisms involved, from policy adoption to actual improvement in child health outcomes.

Author Contribution

Niharika Rustagi: Conceptualisation, formal analysis, writing – original draft, methodology. Sonia Akter: Conceptualisation, methodology, supervision, writing – original draft, and writing– review & editing

Data availability

Data derived from public domain resources.

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Appendices.

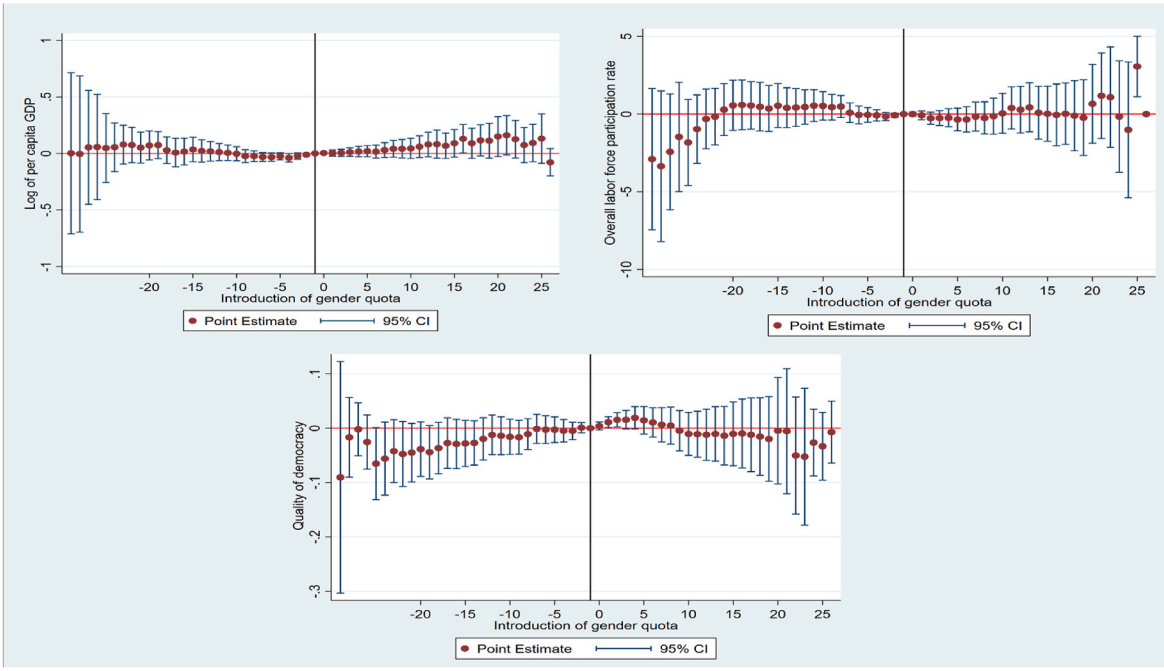
Appendix 1

Impact of women's political representation on children's health outcomes (Fixed effects (FE) regression results)

	(1)	(2)	(3)	(4)	(5)	(6)
	IMR	Female IMR	Male IMR	Neonatal mortality rate	Measles immunization rate	DPT immunization rate
% women in parliament	−0.152	−0.142	−0.162	−0.045	0.020	0.080
	(0.131)	(0.125)	(0.137)	(0.041)	(0.060)	(0.060)
Ln of per capita GDP (in 2010 US$)	−7.332***	−6.650***	−8.001***	−4.107***	−2.600	−2.600
	(2.421)	(2.294)	(2.552)	(1.160)	(3.870)	(3.270)
Democracy index	−9.872*	−9.380*	−10.360*	−3.749*	2.970	3.250
	(5.623)	(5.361)	(5.886)	(2.224)	(4.860)	(4.930)
Constant	111.800***	101.800***	121.400***	58.720***	96.040***	98.620***
	(19.610)	(18.550)	(20.690)	(9.396)	(31.490)	(26.600)
Country FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
R-squared	0.548	0.535	0.559	0.602	0.250	0.230
Observations	4,279	4,279	4,279	4,279	4,265	4,265
Number of countries	162	162	162	162	162	162

Notes: Robust standard errors clustered at the country level in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
IMR, Infant mortality rate.
DPT, diphtheria, pertussis, and tetanus.

Appendix 2. Difference of trends in key socio-economic variables between gender quota adopters and non-adopters



Notes: Point estimates of the lag and lead terms in the event study specification are presented, along with their 95% Confidence intervals. Estimates are conditional on country and year-fixed effects. Robust standard errors are clustered at the country level. The omitted base category is taken as 1 year prior to the reform, indicated by the solid vertical line.

Appendix 3

First election year post-adoption of gender quota as the IV (IVFE regression results)

	(1)	(2)	(3)	(4)
	IMR	Neonatal mortality rate	Measles immunization rate	DPT immunization rate
% women in parliament	−1.420***	−0.550***	0.770***	0.830***
	(0.280)	(0.100)	(0.190)	(0.200)
Country FE	Y	Y	Y	Y
Time FE	Y	Y	Y	Y
R-squared	0.34	0.41	0.14	0.13
Observations	1,833	1,833	1,829	1,829

Notes: IVFE stands for instrument variable fixed effects. Robust standard errors clustered at the country level in parentheses * $p < 0.10$, **$p < 0.05$, ***$p < 0.01$.
IMR, Infant mortality rate.
DPT, diphtheria, pertussis, and tetanus.

Appendix 4

Alternate measurement of democracy index (IVFE Regression results)

	(1)	(2)	(3)	(4)
	IMR	Neonatal mortality rate	Measles immunization rate	DPT immunization rate
% women in parliament	−0.650*	−0.280*	0.580**	0.710**
	(0.390)	(0.150)	(0.270)	(0.300)
Country FE	Y	Y	Y	Y
Time FE	Y	Y	Y	Y
R-squared	0.53	0.56	0.18	0.14
Observations	3,959	3,959	3,948	3,948

Notes: IVFE stands for instrument variable fixed effects. Robust standard errors clustered at the country level in parentheses * $p < 0.10$, **$p < 0.05$, ***$p < 0.01$.
IMR, Infant mortality rate.
DPT, diphtheria, pertussis, and tetanus.

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